AMPLENESS SPREADS OUT FROM FIBRES ALONG A PROPER MORPHISM OF ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The author closes the final step with three named ingredients ('hints'); the composition is the affine-pullback transfer of ampleness from X_i to X. Flagged for the reviewer: this closing step is hint-form rather than fully written out, while everything before it is explicit.

Allowed standard prerequisites (editorial).

- fpqc locality of relative ampleness. Stacks Project, Descent on Spaces, Lemma lemma-descending-property-ample. Reduces to residue fields of points.
- Limits of algebraic spaces. Stacks Project, Limits of Spaces, Lemmas spaces-limits-lemma-proper-limit-of-proper-finite-presentation-noetherian, spaces-limits-lemma-limit-is-affine, spaces-limits-lemma-descend-invertible-modules, spaces-limits-lemma-limit-is-scheme. Supplies the approximation system and its shrinking steps.
- Ampleness at the limit. Stacks Project, Limits, Lemma limits-lemma-limit-ample. Gives $\mathcal{L}_{i, y_i}$ ample for some i.
- Ampleness in a neighbourhood, finite-presentation case. Stacks Project, Divisors on Spaces, Lemma spaces-divisors-lemma-ample-in-neighbourhood. Supplies the open V_i on the model.
- Ampleness under affine pullback and base change. Stacks Project, Morphisms, Lemmas morphisms-lemma-ample-base-change and morphisms-lemma-pullback-ample-tensor-relatively-ample. Named by the author as the ingredients of the final pullback step.

Source reference. Original reference 0D31: sources/stacks-chow/tags.txt, source lines 16401–16401. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0D36: sources/stacks-chow/tags.txt, source lines 16406–16406. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0D3C: sources/stacks-chow/tags.txt, source lines 16412–16412. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0A0X: sources/stacks-chow/tags.txt, source lines 12650–12650. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

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Source reference. Original reference 07SQ: sources/stacks-chow/tags.txt, source lines 9910–9910. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0D2X: sources/stacks-chow/tags.txt, source lines 16397–16397. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 07SR: sources/stacks-chow/tags.txt, source lines 9911–9911. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 09MT: sources/stacks-chow/tags.txt, source lines 12191–12191. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0D3A: sources/stacks-chow/tags.txt, source lines 16410–16410. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0893: sources/stacks-chow/tags.txt, source lines 10483–10483. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0892: sources/stacks-chow/tags.txt, source lines 10482–10482. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a proper morphism of algebraic spaces over S. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module. There exists an open subspace $V \subset Y$ characterized by the following property: A morphism $Y' \to Y$ of algebraic spaces factors through V if and only if the pullback $\mathcal{L}'$ of $\mathcal{L}$ to $X' = Y' \times_Y X$ is ample on X'/Y' (as in Divisors on Spaces, Definition 0D31).*

Proof. Suppose that the lemma holds whenever Y is a scheme. Let U be a scheme and let $U \to Y$ be a surjective étale morphism. Let $R = U \times_Y U$ with projections $t, s : R \to U$. Denote $X_U = U \times_Y X$ and $\mathcal{L}_U$ the pullback. Then we get an open subscheme $V' \subset U$ as in the lemma for $(X_U \to U, \mathcal{L}_U)$. By the functorial characterization we see that $s^{-1}(V') = t^{-1}(V')$. Thus there is an open subspace $V \subset Y$ such that V' is the inverse image of V in U. In particular $V' \to V$ is surjective étale and we conclude that $\mathcal{L}_V$ is ample on X_V/V (Divisors on Spaces, Lemma 0D36). Now, if $Y' \to Y$ is a morphism such that $\mathcal{L}'$ is ample on X'/Y', then $U \times_Y Y' \to Y'$ must factor through V' and we conclude that $Y' \to Y$ factors through V. Hence $V \subset Y$ is as in the statement of the lemma. In this way we reduce to the case dealt with in the next paragraph.

Assume Y is a scheme. Since the question is local on Y we may assume Y is an affine scheme. We will show the following:

(A) If $\mathrm{Spec}(k) \to Y$ is a morphism such that $\mathcal{L}_k$ is ample on X_k/k, then there is an open neighbourhood $V \subset Y$ of the image of $\mathrm{Spec}(k) \to Y$ such that $\mathcal{L}_V$ is ample on X_V/V.

It is clear that (A) implies the truth of the lemma.

Let $X \to Y$, $\mathcal{L}$, $\mathrm{Spec}(k) \to Y$ be as in (A). By Lemma 0D3C we may assume that $k = \kappa(y)$ is the residue field of a point y of Y.

As Y is affine we can find a directed set I and an inverse system of morphisms $X_i \to Y_i$ of algebraic spaces with Y_i of finite presentation over $\mathbf{Z}$, with affine transition morphisms $X_i \to X_{i'}$ and $Y_i \to Y_{i'}$, with $X_i \to Y_i$ proper and of finite presentation,

and such that $X \to Y = \lim(X_i \to Y_i)$. See Limits of Spaces, Lemma 0A0X. After shrinking I we may assume Y_i is an (affine) scheme for all i, see Limits of Spaces, Lemma 07SQ. After shrinking I we can assume we have a compatible system of invertible $\mathcal{O}_{X_i}$-modules $\mathcal{L}_i$ pulling back to $\mathcal{L}$, see Limits of Spaces, Lemma 0D2X. Let $y_i \in Y_i$ be the image of y. Then $\kappa(y) = \operatorname{colim} \kappa(y_i)$. Hence $X_y = \lim X_{i,y_i}$ and after shrinking I we may assume X_{i,y_i} is a scheme for all i, see Limits of Spaces, Lemma 07SR. Hence for some i we have $\mathcal{L}_{i,y_i}$ is ample on X_{i,y_i} by Limits, Lemma 09MT. By Divisors on Spaces, Lemma 0D3A we find an open neighbourhood $V_i \subset Y_i$ of y_i such that $\mathcal{L}_i$ restricted to $f_i^{-1}(V_i)$ is ample relative to V_i. Letting $V \subset Y$ be the inverse image of V_i finishes the proof (hints: use Morphisms, Lemma 0893 and the fact that $X \to Y \times_{Y_i} X_i$ is affine and the fact that the pullback of an ample invertible sheaf by an affine morphism is ample by Morphisms, Lemma 0892). □